

## Proof and Induction WS 4

1. Use a direct proof to prove each of the following statements.
  - a. The sum of two rational numbers is rational.
  - b. If  $n$  is odd, then  $n^2$  is odd.
2. Use a contrapositive proof to prove each of the following statement: If  $x$  is irrational, then  $\sqrt{x}$  is irrational.
3. Use a proof by contradiction to prove each of the following statement: The product of a rational and an irrational number is irrational.
4. Prove each of the following logical equivalence: Let  $n$  be a positive integer.  $n$  is odd if and only if  $7n + 6$  is odd.
5. Prove that a four-digit number is divisible by 9 if and only if the sum of its digits is divisible by 9.
6. Write each mathematical statements in words.
  - a.  $\forall n, m \in \mathbb{Z}, n, m > 0$ , if  $n < m$ , then  $\frac{1}{n} > \frac{1}{m}$
  - b.  $\forall a, b \in \mathbb{R}, a^2 + b^2 \geq 2ab$
  - c.  $\forall p, q \in \mathbb{Q}, p < q, \exists r \in \mathbb{R}: p < r < q$
7. Which of the following is the negation of the statement: " $\forall p \in P$  ( $p$  is of the form  $4m + 1 \Rightarrow p$  can be written as a sum of two squares) " ?
  - (A)  $\forall p \in P$  ( $p$  is of the form  $4m + 1$  and  $p$  cannot be written as a sum of two squares)
  - (B)  $\forall p \in P$  ( $p$  is not of the form  $4m + 1$  and  $p$  can be written as a sum of two squares)
  - (C)  $\forall p \in P$  ( $p$  is not of the form  $4m + 1$  and  $p$  cannot be written as a sum of two squares)
  - (D)  $\forall p \in P$  ( $p$  is of the form  $4m + 1$  and  $p$  cannot be written as a sum of two squares)
8. Let  $n$  be a natural number.
  - a. Show that if  $n$  is composite, then there exists a factor of  $n$  not equal to 1, and at most equal to  $\sqrt{n}$ .
  - b. State the converse of the proposition in part i.
  - c. Write down the contrapositive of the proposition in part i.
9. It is given that  $a$  and  $b$  are positive real numbers. Consider the statement  $\forall a (\forall b, a^{\ln b} = b^{\ln a})$ . Either prove that statement is true or provide a counter example.
10. Let  $x$  be a positive real number, and let  $S$  represent the statement:  $\forall x > 0, x \in \mathbb{R} \setminus \mathbb{Q} \Rightarrow \sqrt{x} \in \mathbb{R} \setminus \mathbb{Q}$ 
  - a. Prove that  $S$  is TRUE.
  - b. Using a counterexample, prove that the converse of  $S$  is FALSE.
11. If  $u_n = 9^{n+1} - 8n - 9$ , show that  $u_{n+1} = 9u_n + 64n + 64$ , and hence show that  $u_n$  is divisible by 64 for  $n \geq 1$ .
12. If  $u_n = 5^n - 4n - 1$ , show that  $u_n > 0$  for  $n \geq 2$ .
13.
  - a. Using the product rule for differentiation, show that for  $n \geq 1, \frac{d}{dx} x^n = nx^{n-1}$
  - b. Using integration by parts, show that for  $n \geq 1, \int x^n dx = \frac{x^{n+1}}{n+1} + c$
14. Use a stronger form of induction to prove the following results for the given sequences.
  - a. If  $T_1 = 3, T_2 = 6$  and  $T_n = 3T_{n-1} - 2T_{n-2} - 1$  for  $n \geq 3$ , then  $T_n = n + 2^n$  for  $n \geq 1$
  - b. If  $T_1 = 8, T_2 = 34$  and  $T_n = 8T_{n-1} - 15T_{n-2}$  for  $n \geq 3$ , then  $T_n = 5^n + 3^n$  for  $n \geq 1$ .
  - c. If  $T_1 = 12, T_2 = 30$  and  $T_n = 5T_{n-1} - 6T_{n-2}$  for  $n \geq 3$ , then  $T_n = 3 \times 2^n + 2 \times 3^n$  for  $n \geq 1$ .
  - d. If  $T_0 = T_1 = 2$  and  $T_n = 2T_{n-1} + T_{n-2}$  for  $n \geq 2$ , then  $T_n = (1 + \sqrt{2})^2 + (1 - \sqrt{2})^n$  for  $n \geq 0$ .
- 15.

a. Prove the identity  $2 \cos A \sin B = \sin(A + B) - \sin(A - B)$ .

b. Hence prove by induction that for all positive integers  $n$ ,  $\frac{1}{2} + \cos \theta + \cos 2\theta + \dots +$

$$\cos(n-1)\theta = \frac{\sin\left(n-\frac{1}{2}\right)\theta}{2 \sin \frac{1}{2}\theta}$$

16. Prove by mathematical induction that for every positive integer  $n$ ,  $\frac{n^5}{5} + \frac{n^3}{3} + \frac{7n}{15}$  is an integer

17. If  $u_1 = 7, u_2 = 29$  and  $u_n = 7u_{n-1} - 10u_{n-2}$  for  $n \geq 3$ , show that  $u_n = 2^n + 5^n$  for  $n \geq 1$ .

18.

a. for  $n \geq 1$

i. Show that  $\frac{1}{n} - \frac{1}{n+1} = \frac{1}{n(n+1)}$  and  $\frac{1}{(n+1)^2} < \frac{1}{n(n+1)}$

ii. Using (i) and the method of induction, prove that:  $S_n = \sum_{r=1}^n \frac{1}{r^2} \leq 2 - \frac{1}{n}$  for  $n \geq 1$

iii. Show  $1.45 \leq S_{99} \leq 1.99$

b. Using the method of induction, prove that for each positive integer  $n$  there are unique positive integers  $a_n$  and  $b_n$  such that  $(1 + \sqrt{2})^n = a_n + \sqrt{2} \cdot b_n$ .

Prove that  $a_n^2 - 2b_n^2 = (-1)^n$

19.

a. Show that  $\frac{\sin\left(\frac{3x}{2}\right) - \sin\left(\frac{x}{2}\right)}{2 \sin\left(\frac{x}{2}\right)} = \cos x$

b. Prove by induction that:  $\cos x + \cos 2x + \cos 3x + \dots + \cos nx = \frac{\sin\left(\frac{2n+1}{2}x\right)}{2 \sin\left(\frac{x}{2}\right)} - \frac{1}{2}$

20. The Fibonacci sequence  $1, 1, 2, 3, 5, 8, \dots$  can be defined as  $u_1 = 1, u_2 = 1, u_{n+2} = u_n + u_{n+1}$  for integers  $n > 0$

a. Use induction to prove that  $u_n < a^n$ , for any  $a > \frac{1+\sqrt{5}}{2}$

b. Assuming that  $\frac{u_{n+1}}{u_n}$  approaches a limit as  $n \rightarrow \infty$ . Show that  $\frac{u_{n+1}}{u_n} \rightarrow \frac{1+\sqrt{5}}{2}$

21. If  $u_1 = 1$  and  $u_n = \frac{1}{2}\left(u_{n-1} + \frac{3}{u_{n-1}}\right)$  for  $n \geq 2$ , show that

a.  $u_n^2 > 3$  for  $n \geq 2$

b.  $u_{n+1} < u_n$  for  $n \geq 2$

22. By considering the binomial expansion of  $(1 + i)^n$  show that:  $1 - \binom{n}{2} + \binom{n}{4} - \binom{n}{6} + \dots = 2^{\frac{n}{2}} \cos \frac{n\pi}{4}$

23. If  $u_1 = 1$  and  $u_n = \sqrt{3u_{n-1}}$  for  $n \geq 2$ , show that

a.  $u_n < 3$  for  $n \geq 1$

b.  $u_{n+1} > u_n$  for  $n \geq 1$

24. If  $u_1 = 1, u_2 = 1$  and  $u_n = u_{n-1} + u_{n-2}$  for  $n \geq 3$ , show that  $u_n = \frac{1}{\sqrt{5}} \left\{ \left( \frac{1+\sqrt{5}}{2} \right)^n - \left( \frac{1-\sqrt{5}}{2} \right)^n \right\}$  for  $n \geq 1$

25. Prove by mathematical induction that  $\sqrt{2 + \sqrt{2 + \sqrt{2 + \dots \sqrt{2}}}} = 2 \cos \frac{\pi}{2^{n+1}}$  for  $n \geq 1$  where there are

$n$  lots of 2's on the left hand side. (e.g.  $\sqrt{2 + \sqrt{2}} = 2 \cos \frac{\pi}{2^{2+1}}$ )

Answer

6. a. For all positive integers  $n$  and  $m$ , if  $n < m$  then  $\frac{1}{n} > \frac{1}{m}$ ; b. For all real numbers  $a$  and  $b$ ,  $a^2 + b^2 \geq 2ab$ ; c. For all rational numbers  $p$  and  $q$  where  $p < q$ , there exists a real number  $r$  such that  $p < r < q$ .

7. D

8. b.  $n$  has a factor less than  $\sqrt{n} \Rightarrow n$  is composite; c. if  $n$  does not have factor  $\leq \sqrt{n}$  then  $n$  is not composite (i.e. prime)